

SEATTLE WEST POINT, WA, USA

WMO: 994350

Lat: 47.6620N

Lon: 122.4353W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	0.9	-8.4	1.8	0.9	-5.5	2.4	2.6	23.3	7.0	18.6	7.9	5.2	10	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	21.3	16.2	20.0	15.8	18.9	15.4	17.2	19.7	16.4	18.8	15.8	17.9	3.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	18.2	15.4	10.9	17.3	14.8	10.5	16.8	48.3	19.9	45.9	18.9	44.1	18.0	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.1	11.8	-2.7	24.9	2.4	1.4	-4.4	25.9	-5.7	26.8	-7.1	27.6	-8.8	28.6		
(5)			-4.3	18.3	2.4	0.7	-6.0	18.8	-7.4	19.2	-8.8	19.7	-10.6	20.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	6.6	6.6	7.9	9.5	12.0	14.1	16.0	16.2	14.7	11.6	8.5	6.4
	DBStd	4.12	2.62	2.40	2.12	1.94	1.99	1.74	1.62	1.31	1.51	1.86	2.52	2.70
	HDD10.0	485	106	96	71	31	5	0	0	0	0	7	56	113
	HDD18.3	2733	362	328	324	264	197	129	77	68	111	208	295	369
	CDD10.0	799	2	1	6	17	66	122	184	191	139	57	11	2
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	7	0	0	0	0	1	1	4	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	5.1	5.9	5.7	5.9	5.4	4.8	4.3	3.9	3.8	4.2	5.1	5.9	5.8
(15) Precipitation	PrecAvg	1015	145	98	109	71	51	42	17	23	43	105	152	165
	PrecMax	1328	296	244	211	143	100	77	44	76	129	212	356	298
	PrecMin	756	72	9	35	10	12	8	0	0	5	21	57	67
	PrecStd	161	58	51	46	38	25	21	12	22	31	52	66	67
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	12.7	15.2	16.9	19.5	22.0	23.9	22.7	20.7	17.3	14.6	13.2
		MCWB	10.4	7.7	10.1	11.2	13.9	15.4	17.7	17.1	15.5	13.8	12.8	11.6
	2%	DB	11.6	11.3	12.8	14.6	17.5	19.6	21.6	20.8	19.0	15.8	13.4	11.6
		MCWB	9.7	8.1	9.3	10.3	13.3	14.8	16.7	16.3	15.2	13.3	12.0	10.0
	5%	DB	10.7	10.3	11.6	13.3	16.1	18.0	20.0	19.8	17.9	15.0	12.7	10.6
		MCWB	9.1	8.0	8.9	10.0	12.7	14.2	16.1	15.9	14.9	12.8	11.2	9.2
	10%	DB	10.0	9.5	10.6	12.2	14.9	16.8	18.8	18.7	16.9	14.1	11.8	9.8
		MCWB	8.5	7.8	8.6	9.5	12.1	13.8	15.6	15.5	14.4	12.3	10.4	8.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	9.9	11.3	12.3	15.2	16.5	18.8	18.4	17.3	15.4	13.5	12.3
		MCDB	12.2	10.9	13.2	15.0	17.5	19.6	21.4	20.8	18.8	15.9	14.1	12.9
	2%	WB	10.4	9.0	10.3	11.2	14.0	15.3	17.4	17.2	16.0	14.1	12.3	10.5
		MCDB	11.1	10.0	11.5	13.2	16.4	17.8	20.1	19.6	17.5	15.2	13.1	11.2
	5%	WB	9.4	8.4	9.4	10.5	13.0	14.7	16.4	16.5	15.2	13.3	11.4	9.5
		MCDB	10.2	9.5	10.7	12.3	15.3	17.0	19.1	18.8	16.9	14.4	12.2	10.4
	10%	WB	8.6	7.8	8.8	9.9	12.3	14.0	15.7	15.8	14.7	12.6	10.5	8.6
		MCDB	9.6	8.9	10.2	11.6	14.4	16.3	18.0	17.9	16.4	13.7	11.5	9.5
(35) Mean Daily Temperature Range	5% DB	MDBR	3.1	3.5	4.1	4.6	4.7	4.8	5.4	5.4	4.4	3.5	3.3	3.1
		MCDBR	3.9	4.2	5.3	6.1	6.4	6.8	7.4	6.9	5.8	4.3	3.9	4.0
	5% WB	MCWBR	2.9	2.7	3.1	3.1	3.4	3.4	3.6	3.3	2.8	2.7	3.1	3.2
		MCWBR	3.7	3.4	4.7	5.3	5.7	6.0	6.6	6.4	5.1	3.9	3.7	3.6
(40) Clear-Sky Solar Irradiance	taub		0.328	0.318	0.327	0.355	0.376	0.364	0.351	0.357	0.345	0.331	0.323	0.330
		taud	2.491	2.554	2.544	2.437	2.396	2.455	2.509	2.500	2.525	2.566	2.537	2.421
	Ebn at Noon		753	856	900	899	887	899	906	887	867	827	756	691
		Edh at Noon	65	74	86	105	114	108	101	98	87	72	61	62
(44) All-Sky Solar Radiation	RadAvg		0.99	1.86	2.82	4.15	5.25	5.75	6.40	5.46	3.96	2.22	1.17	0.81
	RadStd		0.14	0.25	0.33	0.39	0.43	0.58	0.48	0.36	0.36	0.24	0.16	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	-1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page